

Technical International Certifications

Portfolio - GADZIDE Komi Joël

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EXECUTIVE SUMMARY

This portfolio presents **11 international certifications (9 in Programming)** from world-leading universities, representing **300+ hours** of self-directed study validated through **24 technical projects**. Beyond technical expertise, I achieved my Baccalauréat with **Honors (16/20)** and the BEPC with a record-breaking **19.50/20**, demonstrating a unique ability to maintain academic excellence while managing significant personal and family responsibilities. The combination of theoretical knowledge from Duke University, University of Michigan, University of Illinois, and UC Irvine with practical implementation across multiple programming domains demonstrates exceptional readiness for advanced software engineering education and meaningful contributions to International technological cooperation.

Key Achievement: All certifications completed independently while maintaining Baccalauréat(High Secondary School Diploma) ,Mention Très Bien(With Honor), Grade :16/20 ; demonstrating exceptional self-motivation and technical aptitude.

Strategic Application: These skills directly support my goal to develop autonomous systems (Multi-purpose drones and Robots, smart home solutions) for for Africa's technological advancement through Sino-African cooperation.

Certifications Summary

#	Certificate	Institution	Completed	Hours	Key Skills	GitHub Projects
1	Programming Foundations with JavaScript, HTML & CSS	Duke University (Coursera)	2025	32	JavaScript, HTML/CSS, DOM manipulation, web development	Plant Store Landing Page
2	Programming Fundamentals	Duke University (Coursera)	2025	18	Core programming logic, computational thinking, problem-solving	Tic-Tac-Toe GUI Game (Python Tkinter)
3	Programming for Everybody (Python)	University of Michigan (Coursera)	2025	18	Python fundamentals, data types, loops, file I/O	Numerical Analysis Notebooks

						(Multiple)
4	Object-Oriented Data Structures in C++	University of Illinois (Coursera)	2025	21	C++ OOP, data structures (linked lists, trees, stacks), memory management	C Linked List Operations , C++ Banking System
5	Introduction to IoT & Embedded Systems	University of California Irvine (Coursera)	2025	11	IoT architecture, embedded hardware, system design, protocols	Arduino Serial LED Projects (2 projects)
6	Java Programming: Solving Problems with Software	Duke University (Coursera)	2025	18	Java fundamentals, OOP principles, algorithms, problem-solving	Library Management System , Pokemon Battle System , Store Project
7	Mastering Python Sequences and Strings	EDUCBA (Coursera)	2025	7	Python data structures optimization, string manipulation, performance	Applied Numerical Methods , Matrix Analysis
8	Machine Learning with Python	FreeCodeCamp	2025	100	Supervised/unsupervised learning, scikit-learn, neural networks, model evaluation	ML Implementations , Data Analysis Projects
9	The Arduino Platform and C Programming	University of California Irvine (Coursera)	2025	12	C Programming, Embedded Software, Microcontrollers (Arduino), Hardware Interfacing, Computational Thinking	C Linked List Operations

Total: 220+ documented hours (additional 80+ hours in project development and debugging)

Note: All certificates are verified with unique platform IDs and available for inspection upon request.

Technical Projects Breakdown

Programming Fundamentals & Web (3 projects)

- **Plant Store Landing Page** (HTML/CSS/JavaScript) – Responsive web design
- **Tic-Tac-Toe GUI** (Python Tkinter) – Event-driven programming
- **Multiple web components** – DOM manipulation, interactivity

Object-Oriented Programming (6+ projects)

Java:

- **Library Management System** – Complete CRUD with inheritance
- **Pokemon Battle System** – Abstract classes, polymorphism
- **Movie Recommendation System** – Data analysis algorithms
- **Store Management** – Inventory and transaction handling

C++:

- **Banking Management System** – Advanced OOP with file I/O
- **Scientific Calculator** – Expression parsing, data structures
- **Mini Compiler** – Lexer, parser, interpreter implementation
- **Snake Game** – Real-time game loop, collision detection

Data Structures & Algorithms and The Arduino Platform & C Programming (4+ projects)

- **C Linked List Operations** – Manual memory management
- **Network Optimization (Dijkstra's Algorithm)** – Graph theory
- **Graph Representation & Adjacency** – Multiple implementations
- **Custom data structures** – Used across all C++ projects

Python Scientific Computing (10+ projects)

- **Numerical Analysis with SymPy** – Symbolic mathematics
- **Applied Kinematics & Integration** – Physics simulations
- **Particle Kinematics & Arc Length** – Computational physics
- **Matrix Analysis & Linear Systems** – Linear algebra
- **Multivariable Function Visualization** – 3D graphing
- **Applied Numerical Methods** – Calculus and algebra
- **Multiple Jupyter notebooks** – Scientific computing workflows

Embedded Systems & IoT (2 projects)

- **Arduino Serial LED Control (Project 1)** – Serial communication
- **Arduino Serial LED Control with Web Interface (Project 2)** – IoT integration

Game Development (2 projects)

- **Endless Car Runner** (Python Pygame) – Game mechanics, sprites
- **Snake Game** (C++ Console) – Real-time input, collision detection

GitHub Stats: 20+ repositories | 6 programming languages | 100+ commits

Certification Details: Course Requirements & Practical Validation

Each certification required specific technical deliverables and was validated through rigorous projects demonstrating mastery beyond theoretical knowledge.

Domain 1: Software Engineering & Object-Oriented Programming

1- Java Programming: Solving Problems with Software (Duke University)

- **Requirement:** Design and implement complete systems using OOP principles (inheritance, polymorphism, encapsulation)
- **Score:** 88.31% (Passed with Honors)
- **Validation:** Mastery of complex Data Structures (HashSet, HashMap) and encryption algorithms.

- **Extended Applications(Post-Completion) :** Developed three production-quality Java systems:
- **Library Management System** – Demonstrates class hierarchies and data persistence
- **Pokemon Battle System** – Abstract base classes with type-based polymorphism
- **Store Management** – Real-world business logic with CRUD operations
- **GitHub:** All source code fully documented and available

2- Object-Oriented Data Structures in C++ (University of Illinois)

- **Requirement:** Implement complex data structures (linked lists, trees, hash tables) with manual memory management
- **Score:** 96%
- **Validation:** Data structures (lists, trees) using C++ Object-Oriented Programming (OOP)
- **Extended Applications(Post-Completion) :**
- **C Linked List Operations** – Manual pointer manipulation, memory allocation
- **C++ Banking System** – STL containers, smart pointers, file I/O
- **C++ Projects** – Applied DS concepts across multiple projects
- **Skills Demonstrated:** Low-level memory management, STL mastery, performance optimization

Domain 2: Programming Fundamentals & Logic

3- Programming Fundamentals (Duke University)

- **Score:** 97.50% (Honors)
- **Requirement:** Successfully complete computational thinking challenges and algorithmic mini-projects
- **Validation:** Computational Thinking challenges and mini-projects requiring the application of algorithms.
- **Extended Applications(Post-Completion) :** Tic-Tac-Toe GUI game demonstrating structured problem-solving approach
- **Impact:** Foundation for all subsequent algorithmic work

4- Programming for Everybody – Python (University of Michigan)

- **Score:** 98.25% (Top Performance)
- **Requirement:** File processing, data parsing, and analysis exercises
- **Validation :** File processing and basic data analysis (Data Parsing)
- **Extended Applications(Post-Completion) :** Multiple Python projects including:
- Numerical analysis notebooks
- Data manipulation workflows
- Scientific computing applications
- **Impact:** Python became primary language for scientific computing

5- Mastering Python Sequences and Strings (EDUCBA)

- **Requirement:** Deep understanding of Python data structures for performance optimization
- **Score:** 87.50%
- **Validation:**

- Optimized data handling across all Python projects
- Python sequential objects (lists, tuples, dictionaries, sets) to ensure high performance and efficiency of data structures
- **Application:** Used in every Python notebook for efficient data preparation and algorithmic implementations

6- The Arduino Platform and C Programming (UC Irvine)

- **Score:** 97.50% (Honors)
- **Requirement:** Successfully complete computational thinking challenges and algorithmic mini-projects
- **Validation:** Computational Thinking challenges and mini-projects requiring the application of algorithms.
- **Extended Applications(Post-Completion) :** Tic-Tac-Toe GUI game demonstrating structured problem-solving approach.
- **Impact:** Foundation for all subsequent algorithmic work

Domain 3: Artificial Intelligence & Machine Learning

7- Machine Learning with Python (FreeCodeCamp)

- **Requirement:** Complete 5 mandatory ML projects including classification, regression, and neural networks
- **Hours:** 100 (most intensive certification)
- **Validation:**
- Implemented supervised/unsupervised learning algorithms
- Built, trained, and validated multiple ML models
- Applied to real datasets with performance evaluation
- **Relevance:** Foundational for autonomous systems (drones, smart homes) requiring AI capabilities

Domain 4: Web Development & Computational Thinking,

8- Programming Foundations with JS, HTML and CSS (DUKE University)

- **Requirement:** Design and implement foundational web applications, focusing on event-driven programming, image processing, and algorithms for interactive user interfaces.
- **Score:** 86.42%
- **Validation:**
- Computational Thinking: Proven capacity to design programming solutions using core concepts (functions, loops, conditionals).
- Algorithms & Image Analysis: Successful implementation of algorithms for dynamic image manipulation, including creating filters (e.g., grayscale, green screen effect) via JavaScript pixel processing.
- Interactive Design: Mastery of Event-Driven Programming, connecting user actions to dynamic responses on the webpage.

- **Web Structure (UI/UX):** Competence in building structured web pages using proper HTML elements (paragraphs, links, images) and applying effective CSS styling (using IDs and classes).
- **Skills:** JavaScript, HTML/CSS, DOM manipulation, User Interface (UI) Design, web development
- **Extended Applications(Post-Completion) :** Plant Store Landing Page (A project demonstrating competence in responsive front-end design, directly utilizing the HTML/CSS/JavaScript skills acquired.)

Domain 5: Embedded Systems & Internet of Things

9- Introduction to IoT & Embedded Systems (UC Irvine)

- **Requirement:** Design complete embedded system architecture and analyze hardware-software trade-offs
- **Score:** 91%
- **Validation:**
 - Theoretical design of IoT systems
 - Understanding of protocols (MQTT, HTTP, I2C)
 - Real-time sensor integration
 - Applied in Arduino projects
- **Skills:** Debugging embedded systems, interrupt handling, hardware constraints
- **Real-World Extension:** Enhanced by intensive 1-month phone repair experience (hardware troubleshooting, component-level understanding)

LEADERSHIP & COMMUNITY IMPACT

My technical journey is driven by a deep commitment to social progress and community service:

- **Ashinaga Leadership Finalist (2025):** Recognized internationally by the Ashinaga Foundation for exceptional academic potential and leadership qualities as a future African leader.
- **Plan International Togo (2019-2020):** Volunteered in rural areas (Notsè) for menstrual hygiene awareness campaigns, demonstrating early social responsibility.
- **Internet Society (ISOC) Togo:** Active participant in digital inclusion discussions to leverage technology for social empowerment.
- **Technical Mentorship:** Sharing programming resources and hardware troubleshooting expertise (mobile repair) with local youth to foster a community of innovators.

Skills Matrix

Skill Category	Proficiency Level	Certifications	Projects
Object-Oriented Programming	Advanced	1	6+

Python Scientific Computing	Advanced	2	10+
Data Structures & Algorithms	Intermediate	3	5+
Embedded Systems & IoT	Intermediate	1	2+
Machine Learning	Intermediate	1	Multiple
Web Development	Intermediate	1	2+
Game Development	Intermediate	-	3

Impact & Strategic Relevance

Academic Preparedness

These certifications provide comprehensive preparation for university-level computer science and software engineering programs:

- **Theoretical Foundation:** Strong understanding of CS fundamentals, algorithms, and system design
- **Practical Skills:** 20+ completed projects demonstrating ability to build production-quality software
- **Self-Learning Ability:** 300+ hours of independent study proves motivation and discipline for rigorous academic programs

Research Potential

Skills directly applicable to cutting-edge research areas:

- **Robotics:** Embedded systems + algorithms + AI/ML
- **IoT Solutions:** Network programming + hardware integration
- **Smart Systems:** ML models + embedded deployment + real-time processing

Development Impact for Togo

Technical foundation to address local challenges:

- **Agricultural Technology:** ML for crop monitoring + IoT sensors + drone control systems
- **Smart Infrastructure:** Embedded systems for smart homes adapted to African power/connectivity constraints
- **Education:** Share knowledge through workshops, mentor young programmers

Global Technology Adaptation & Innovation Potential

This technical foundation positions me to facilitate meaningful technological cooperation:

- **Adaptation Capability:** My experience in resource-constrained computing and embedded systems prepares me to adapt high-end global technologies to diverse local contexts, ensuring they are functional and sustainable.

- **Knowledge Transfer:** A comprehensive understanding of both theoretical Computer Science and practical Software Engineering enables me to act as a catalyst for effective technology transfer between international research and local industries.
- **Innovation Bridge:** I possess the analytical ability to identify emerging global technologies with high potential for regional adaptation, deployment, and socio-economic development.

Verification & Authenticity

- ✓ All certifications include unique verification IDs
- ✓ Digital certificates available upon request (on : <https://github.com/gadzide-joel/Portfolio-projects/tree/main/CERTIFICATIONS>)
- ✓ Coursera profiles publicly viewable
- ✓ GitHub repository is public: <https://github.com/gadzide-joel/Portfolio-projects>
- ✓ Course completion emails and grade reports available

Attestation: I, GADZIDE Komi Joël, certify that all information provided is accurate and verifiable. I completed each certification independently through legitimate online platforms.

Contact Information

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STRATEGIC VALUE AND GLOBAL COMPETENCE

This portfolio demonstrates not just course completion, but practical mastery through 20+ implemented projects spanning web development, embedded systems, machine learning, and scientific computing. The combination of theoretical knowledge from prestigious international institutions and hands-on project experience demonstrates:

1. Exceptional Readiness for advanced software engineering studies in high-standard academic environments.
2. Proven Capacity to bridge complex theoretical knowledge with practical, real-world implementation.
3. Technical Versatility in emerging fields such as AI, IoT, and large-scale systems, which are driving global digital transformation.
4. Tangible Potential to contribute to international technological cooperation and innovation-led development.

FUTURE INNOVATION ROADMAP: SMART SOLUTIONS FOR AFRICA

Development Vision for Togo:

1. **Smart Agriculture:** Designing autonomous drones and robots to automate rural labor and attract youth back to the agricultural sector.
2. **Smart Security:** Developing biometric and renewable energy home solutions for high-security environments, even in low-connectivity areas.
3. **Center of Excellence:** Establishing a technical hub in Togo to incubate local solutions and provide high-level IT training.

CONCLUSION

This portfolio provides concrete evidence of technical capability, self-learning discipline, and research potential essential qualities for success in any rigorous international academic program. The quantifiable achievements (300+ hours, 9 certifications, 20+ projects) demonstrate not just ambition but proven ability to execute complex technical work independently.

The combination of theoretical knowledge from world-class institutions and practical implementation across multiple programming domains creates a unique profile. This trajectory is perfectly aligned with a commitment to applied technological innovation and the pursuit of excellence in the global software engineering landscape.